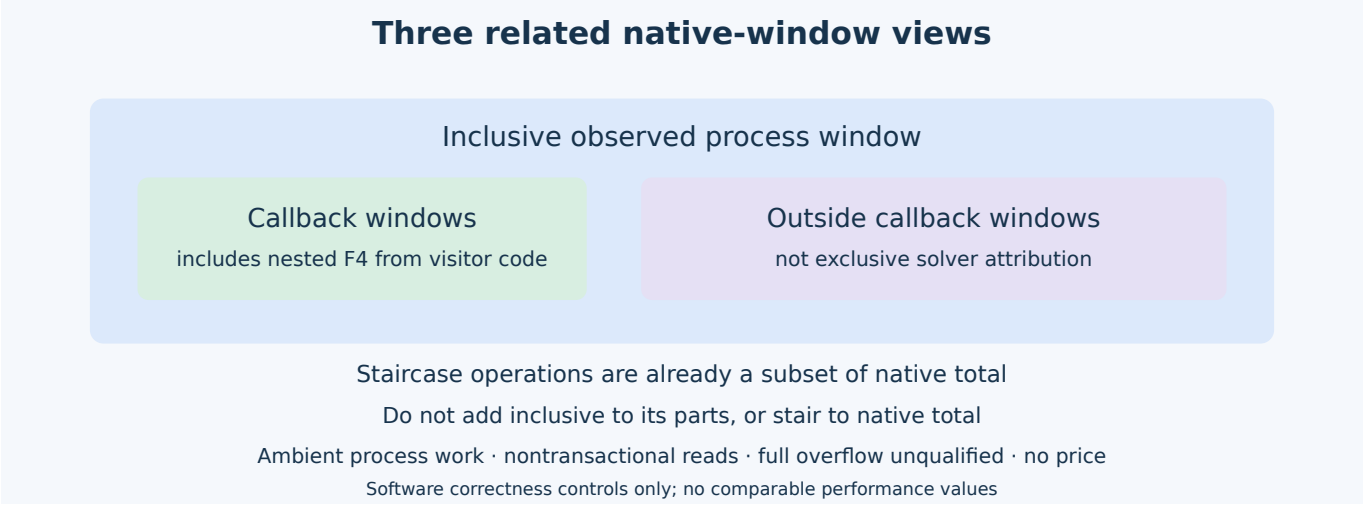
G2: emit observed native F4 windows

2026-11-06 · source cd76e762319d7d7f0cb1239c157e8f81e753d68b



New data, bounded attribution

The oracle emits previously unreported native F4 total/stair observations around the unchanged solve_visit and each actual callback. Checked arithmetic separates observed callback windows from outside windows, retains Result-refusal observations with a caller-held owner, and rejects reused owners without overwriting prior data. Replay checks metadata arithmetic and visitor/attempt/callback consistency. Older records without the optional field remain readable. `[pcc:unverified]`

The provider uses process-wide relaxed atomic counters for incomplete native classes. Concurrent work can contaminate windows; separate loads are not a transaction. Valid-looking differences do not establish isolation or rule out a full wrap. Outside callback windows are not exclusive solver costs. Stair is already included in native total. Inclusive is another view of its parts, not an additional charge. Price remains null, complete coverage false and both provider-overflow and solver-attribution qualifications false.

Fresh source and meaningful controls

Phase	Observed result
Oracle example suite	36 pass, 0 fail/ignored
New native / replay controls	3 / 1 included above
Formatting / CLI build	0 / 0
Actual checkpoint refusal	expected1 + message
Final scoped Cargo lint / supervisor	101 / 101: FAIL

Claim PCC-20261106-496201 is unsigned/unverified. Exact22object/global-clean guards, small phase captures/terminals,2freshbinaries/stablecopies and8unchangedbinary/oldraw/formal bindings are retained. No shared-core algorithm change or redundant old library/process/Lean run.
`[pcc:unverified]`

Consistency controls and remaining qualification

Four new controls

1. A real nested F4 call inside a visitor produces positive callback-native work. Outside-plus-callback observations reconcile to inclusive; tampered outside counts fail metadata checks.
2. Invalid input retains a rejected Result state with zero callbacks. Not-called data is distinct, and reusing the same owner cannot overwrite it.
3. Synthetic backwards, inconsistent-subset and overflow differences become unavailable rather than zero or wrapped exact values. This does not qualify every underlying provider wrap/race.
4. Replay binds the native attempted state to visitor presence and callback diagnostics. Older records without the field still replay for correctness; missing cost data is not free. [pcc:unverified]

A producer can fabricate consistent metadata. The new checks are not independent execution proof or complete operation accounting. Returned means Result::Ok, including incomplete/capped solver reports, not successful scalar recovery. A held owner keeps Result-refusal observations, but the upper producer run API can still discard its local ledger on Err. No durable failed-process/crash/resume claim follows.

Root arithmetic remains an explicit gap

The solver-tree root multiplication field and callback root multiplication field are two executed phases, not duplicate views. Both are partial. Source review finds linear and p2 shortcuts in poly_roots_fp_counted_until that omit the supplied multiplication accumulator; inverse/addition/control work is also incomplete. The p2 shortcut is outside current PKM p>11 inputs; linear cases remain relevant. This step leaves the core unchanged and records the gap instead of claiming full root coverage. A later core change requires new source-bound affected validation. [pcc:unverified]

Legacy solve_excluding_callbacks clocks subtract named callback phase timers, not all wrapper/observer work. No denominator, timing comparison, complete price or GAE result was introduced. The field_ops_total provider’s configured thread/process behavior remains unchanged. Full symbolic/substitution/matrix/logical/root/native work, guards, metadata/serialization, memory, whole-process/journal and qualified runtime/resume remain open.

Gates and preserved evidence

Final strict Cargo lint remains101 on inherited unused solve_one_with_trace and custody bool_comparison; final example lint is unqualified. Older clock failures/full-suite cap/legacy PCC failures remain at ownsource. The formal/gate/Mathlib objects retain their earlier18declaration/0trusted scope and kernel replay against imports, not –fresh. No new Rust/polynomial/complete-cost bridge or theorem is claimed. [pcc:unverified]

Original G1–G3/G6 partial,G4/G5 unrun,scientific panels0,allmetricsnull,comparison_admissible=false. Full-panel card remains blocked/unfunded/unimplemented. No checkpoint/protocol/runtime or funding/lease/IAM/quota/pool/access change. Frozen PCC fork requires a sanctioned contract; exact-head gate,size,artifacts/reviews/SHAguard remain necessary. No frozen resolution or waiver was attempted.

Sources and artifacts: G2_NATIVE_WINDOWS_20261106.md; actual F4/root source and prior G2 audit; official Rust relaxed-atomic documentation; validation/native-windows-20261106/{start,receipt,delivery-review}.json. Diagram is software ownership, not a performance graph. Canonical scientific/curve/performance views and all older report bytes remain unchanged.